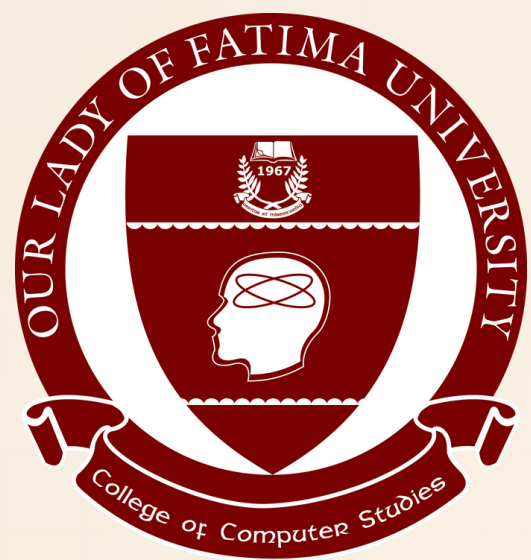




Module 2

History of Computer

ITCL112 - INTRODUCTION TO COMPUTING W/ LAB

INTRODUCTION



The history of computers traces the evolution of technology from simple calculation tools to powerful digital machines. Early devices like the abacus and mechanical calculators paved the way for modern computing. With the invention of electronic computers in the 20th century, machines became faster, smaller, and more intelligent, shaping today's digital age.

MODULE 2

- 1. Understand how computers evolved across generations.**
- 2. Identify the key technology used in each generation**
- 3. Recognize the impact of each generation**
- 4. Types and classifications of computers**
- 5. Information Technology vs ICT vs Computer Science**



GENERATION OF COMPUTER

A generation of computers refers to a specific period in computer history that is marked by a major technological breakthrough in hardware and software. Each generation led to faster, smaller, and more powerful computers.

-  **1ST GENERATION (1940 – 1956)**
-  **2ND GENERATION (1956 – 1963)**
-  **3RD GENERATION (1964 – 1971)**
-  **4TH GENERATION (1971 – PRESENT)**
-  **5TH GENERATION (PRESENT - FUTURE)**

FIRST GENERATION OF COMPUTER



⚙️ 1ST GENERATION (1940 – 1956)

The major hardware features are vacuum tubes and magnetic drums and the processing speed is 1,000 instructions per second

These computers were very heavy and really large. They are not reliable, programming on them was a tedious task as they used low-level programming language and used no Operating System. First-generation computers were used for calculation, storage, and control purposes. Punch cards were used to improve the information for external storage.

FIRST GENERATION OF COMPUTER

Vacuum Tubes



ComputerHope.com

⚙️ 1ST GENERATION (1940 – 1956)

A vacuum tube, also sometimes called an electron tube or valve, is a device that controls the flow of electrical current within a sealed glass enclosure where most of the air has been removed (creating a "vacuum"). By applying electrical signals to different parts of the tube (called electrodes), the flow of electrons can be controlled. In first-generation computers, vacuum tubes acted as electronic switches, enabling the machines to perform calculations.

1. ENIAC (Electronic Numerical Integrator and Computer - 1945)

What it was: The very first general-purpose electronic computer. It was initially built for military calculations. Imagine a machine with around 17,000 vacuum tubes that filled an entire room!

Why it was important: ENIAC showed the world the potential of programmable digital computers. It was a huge leap forward.

2. UNIVAC I (Universal Automatic Computer I 1951)

What it was: The first computer that was made and sold commercially. It was designed for both business and scientific uses.

Why it was important: UNIVAC I marked the beginning of computers being used in everyday business operations.

FIRST GENERATION OF COMPUTER



⚙️ 1ST GENERATION (1940 – 1956)

3. IBM 650 MAGNETIC DRUM DATA-PROCESSING MACHINE

It was the first mass-produced computer in the world. Almost 2,000 systems were produced, in 1962.

4. IBM Harvard Mark I (1944)

What it was: A large computer that used both electrical and mechanical parts. It was used for military and scientific calculations.

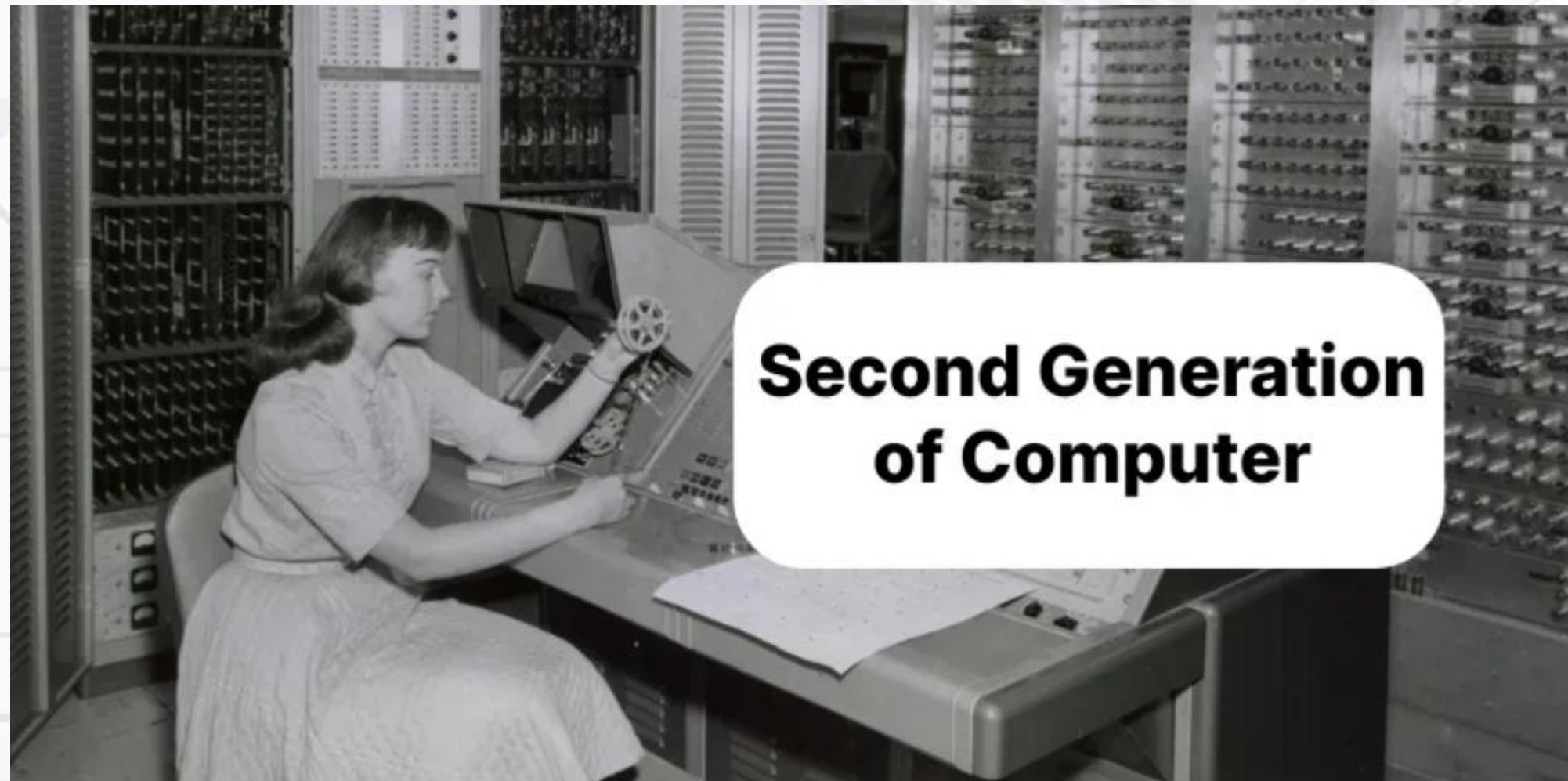
Why it was important: The Mark I was an important step in the development of computing, particularly for scientific and defense applications.

5. ABC (Atanasoff-Berry Computer) (1937-1942)

What it was: One of the earliest attempts at building an electronic digital computer. It was designed to solve algebraic equations.

Why it was important: The ABC introduced some key ideas that would be used in later computer

SECOND GENERATION OF COMPUTER



2ND GENERATION (1956 – 1963)

The major hardware features are transistors and magnetic cores. Its processing speed is 1,000,000 instructions per second.

Second-generation computers used the technology of transistors rather than bulky vacuum tubes. Another feature was the core storage. A transistor may be a device composed of semiconductor material that amplifies a sign or opens or closes a circuit.

SECOND GENERATION OF COMPUTER



2ND GENERATION (1956 – 1963)

TRADIC

1. Transistor Digital Computer

- Built in 1954 by Bell Labs.
- It was the first transistorized computer in the U.S.
- It replaced vacuum tubes with transistors, making it more reliable and energy efficient.

2. Transistorized Airborne Digital Computer

- A later adaptation of TRADIC designed for aircraft use.
- Built to be compact, lightweight, and durable enough for airborne applications.

IBM 1400 SERIES

Stored information in magnetic cores as variable-length character strings separated on the left by a special bit, called a "wordmark," and on the right by a "record mark." Arithmetic was performed digit-by-digit.

THIRD GENERATION OF COMPUTER



3RD GENERATION (1964 – 1971)

The major hardware features are integrated circuits (IC) or “chips” and the processing speed is 10,000,000 instructions per second.

During the third generation, technology envisaged a shift from huge transistors to integrated circuits, also referred to as ICs. Here, a variety of transistors were placed on silicon chips, called semiconductors. The most important feature of this era’s computers was speed and reliability. IC was made from silicon, also called silicon chips.

THIRD GENERATION OF COMPUTER



3RD GENERATION (1964 – 1971)

IBM SYSTEM 360

First general-purpose machine used in science and business.

- IBM 360 Series (1964) – One of the most famous and widely used.
- IBM 370 Series
- Honeywell 6000 series
- PDP-8 (Digital Equipment Corporation)
- ICL 2900 series

The third generation of computers used integrated circuits, which revolutionized computing by making machines smaller, faster, cheaper, and more reliable, paving the way for modern computers.



FOURTH GENERATION OF COMPUTER



4TH GENERATION (1971 – PRESENT)

The major hardware features are microprocessors or large-scale integrated circuits. The minimum processing speed is 100,000,000 instructions per second.

The advantage of this technology is that one microprocessor can contain all the circuits required to perform arithmetic, logic, and control functions on one chip. LSI placed thousands of transistors onto a single chip.

FOURTH GENERATION OF COMPUTER



4TH GENERATION (1971 – PRESENT)

The computers using microchips were called microcomputers. This generation provided even smaller-sized computers with larger capacities. That's not enough, then Very Large Scale Integrated (VLSI) circuits replaced LSI circuits. The Intel 4004 chip, developed in 1971, located all the components of the pc from the central processing unit and memory to input/ output controls on one chip and allowed the dimensions to reduce drastically. VLSI placed several hundred thousand transistors on a single silicon chip. This silicon chip is known as the microprocessor.

- IBM 5100 (first portable computer, 1975).
- Apple II (1977).
- IBM PC (1981, early personal computer).
- Commodore 64 (1982, very popular home computer).
- Intel 4004 (first commercial microprocessor, 1971).

FIFTH GENERATION OF COMPUTER



5TH GENERATION (PRESENT - FUTURE)

The use of parallel processing and superconductors is helping to make artificial intelligence a reality. The goal is to develop devices that respond to natural language input and are capable of learning and self-organization. There are some applications, such as voice recognition, that are being used today.

Fifth Generation Computers = AI + Supercomputing + Quantum Computing + Smart Devices

FIFTH GENERATION OF COMPUTER



5TH GENERATION (PRESENT - FUTURE)

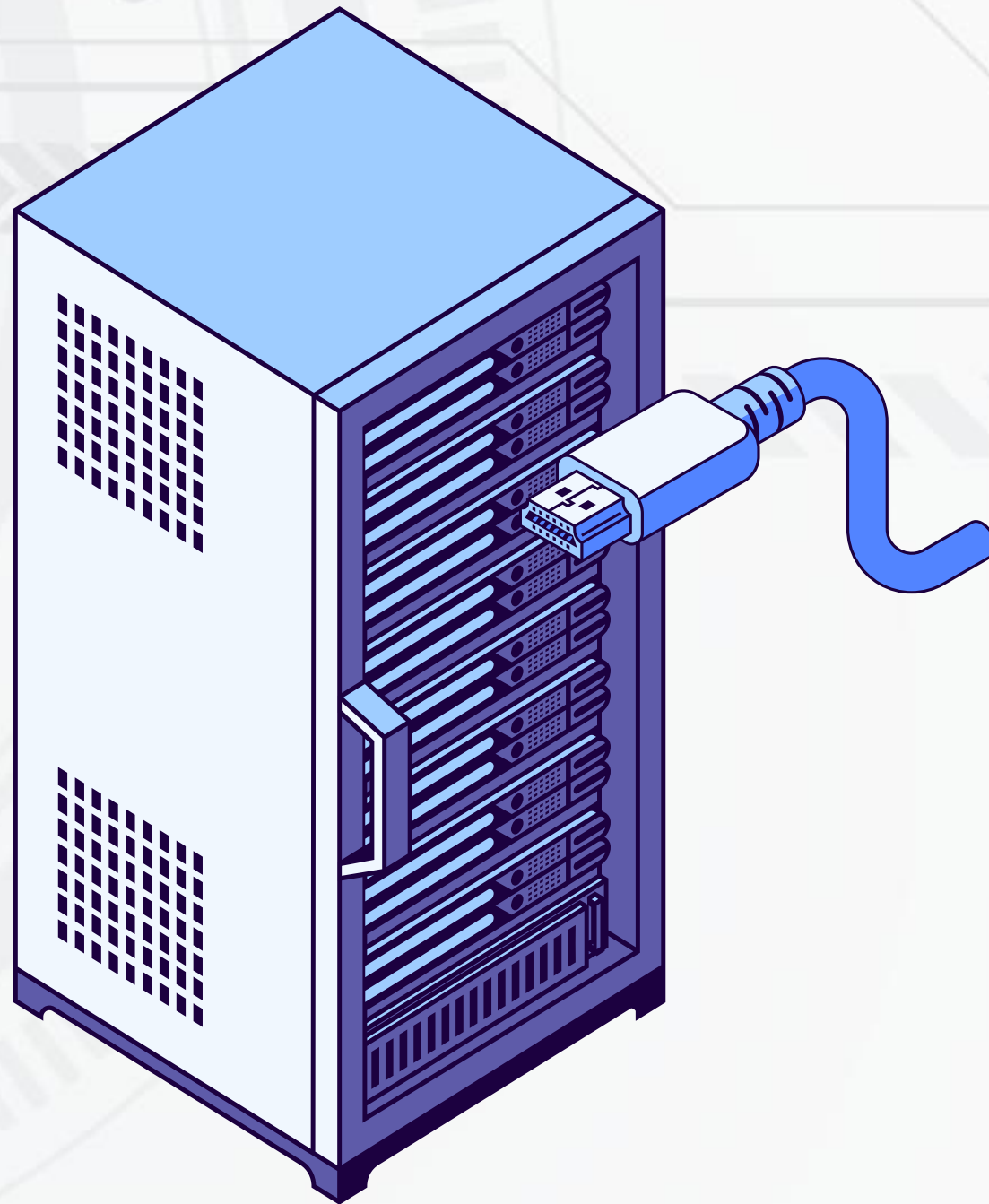
Key Features:

- Based on Artificial Intelligence (AI) and Machine Learning
- Use of parallel processing and superconductors
- Natural language processing (NLP) – computers can understand human languages
- Use of Ubiquitous computing (embedded in everyday devices)
- Faster and more powerful with quantum computing research
- User-friendly interfaces and voice recognition

Examples:

- AI-based systems like IBM Watson
- Voice assistants like Siri, Alexa, and Google Assistant
- Supercomputers (e.g., Fugaku in Japan, Summit in the USA)
- Robotics and Expert Systems used in medicine, engineering, and customer service
- Quantum Computers (Google's Sycamore, IBM Q System One)

TYPES AND CLASSIFICATIONS OF COMPUTERS



1.

PURPOSE

2.

TYPE

3.

SIZE

GENERAL PURPOSE COMPUTER



LESSON 2 : TYPES AND CLASSIFICATIONS OF COMPUTERS

A general-purpose computer is a type of computer designed to perform a wide variety of tasks. Unlike special-purpose computers (which are made for a single or limited function, like ATMs or washing machine controllers), general-purpose computers can run different types of programs and applications depending on the user's needs.

Examples of General-Purpose Computers:

1. Desktop computers
2. Laptop computers
3. Tablets and smartphones

SPECIAL PURPOSE COMPUTER



LESSON 2 : TYPES AND CLASSIFICATIONS OF COMPUTERS

A special-purpose computer is a type of computer designed to perform a specific task or a limited set of tasks efficiently. Unlike general-purpose computers (like desktops or laptops) that can run many kinds of programs, special-purpose computers are optimized for one job and usually cannot be reprogrammed for other uses.

Characteristics of Special-Purpose Computers:

- **Built for a specific application.**
- **Highly efficient and fast in performing that task.**
- **Limited flexibility (cannot easily perform other functions).**
- **Often embedded in larger systems or devices.**

Examples:

- 1. ATMs (Automated Teller Machines)**
- 2. Washing machine controller**
- 3. Traffic light controller**
- 4. Video game consoles**
- 5. Aircraft autopilot system**

ANALOG COMPUTER



LESSON 2 : TYPES AND CLASSIFICATIONS OF COMPUTERS

An analog computer is a type of computer that works by measuring continuous data rather than using discrete values like digital computers. Instead of using binary numbers (0s and 1s), it represents information with physical quantities such as voltage, current, mechanical motion, fluid pressure, or temperature.

It is especially useful for solving complex mathematical equations in real time, such as differential equations, by modeling them physically.

Examples of Analog Computers:

1. Slide Rule
2. Differential Analyzer
3. Planimeter
4. Speedometer in cars
5. Analog flight simulators

DIGITAL COMPUTER



LESSON 2 : TYPES AND CLASSIFICATIONS OF COMPUTERS

A digital computer is a type of computer that processes data in binary form (0s and 1s). It performs calculations, logical operations, and data storage using digital signals. Digital computers are widely used because they are fast, accurate, and can handle a wide range of tasks, from simple calculations to complex simulations.

Key Features of Digital Computers:

- Process data in discrete (digital) form
- High speed and accuracy
- Can store large amounts of data
- Programmable to perform different tasks
- Used in almost all modern applications (business, education, science, communication, etc.)

Examples of Digital Computers:

1. Desktop Computers (e.g., Windows PC, iMac)
2. Laptops (e.g., Dell, HP, MacBook)
3. Smartphones (e.g., iPhone, Samsung Galaxy)
4. Tablets (e.g., iPad, Samsung Tab)
5. Supercomputers (e.g., Fugaku in Japan)

HYBRID COMPUTER



LESSON 2 : TYPES AND CLASSIFICATIONS OF COMPUTERS

A Hybrid Computer is a type of computer that combines the features of both analog and digital computers.

- **Analog part – used for solving complex mathematical equations, continuous data measurement, and simulations (e.g., temperature, speed, or pressure).**
- **Digital part – used for logical operations, data processing, storage, and control.**

This makes hybrid computers very powerful because they can process both continuous data (analog) and discrete data (digital) at the same time.

Examples of Hybrid Computers

- 1. Medical Equipment**
- 2. Automatic Control Systems**
- 3. Gasoline Pump Machines**
- 4. Weather Forecasting Systems**

- 1. SUPERCOMPUTER**
- 2. MAINFRAME
COMPUTER**
- 3. MINI-COMPUTER**
- 4. MICRO-COMPUTER**



LESSON 2 : TYPES AND CLASSIFICATIONS OF COMPUTERS

1. Supercomputer

Definition: The most powerful type of computer, capable of processing trillions of calculations per second.

Purpose: Used for complex tasks like weather forecasting, nuclear simulations, space research, and AI.

2. Mainframe Computer

Definition: Large and powerful systems designed to handle and process massive amounts of data simultaneously.

Purpose: Commonly used in banking, government, and large enterprises for transaction processing, census, and airline reservations.

3. Minicomputer

Definition: A medium-sized computer, more powerful than microcomputers but less powerful than mainframes.

Purpose: Used in small to medium businesses for database management, manufacturing process control, and business applications.

4. Microcomputer

Definition: The smallest type of computer, designed for individual use.

Purpose: Used for everyday tasks like word processing, internet browsing, gaming, and office work.

INFORMATION
TECHNOLOGY

ICT

COMPUTER SCIENCE

 LESSON 3 : INFORMATION TECHNOLOGY VS ICT VS
COMPUTER SCIENCE

Term	Definition	Focus Area	Examples
Information Technology (IT)	Use of computers and software to manage data	Systems, hardware, networks, software for business solutions	IT support, network admin
Information and Communications Technology (ICT)	Broader scope that includes IT + communication tools	Integration of telecom, broadcasting, internet, and IT	E-learning, online banking
Computer Science (CS)	Academic discipline focusing on computation and algorithms	Theoretical and technical foundations of computing	Programming, AI, data structures